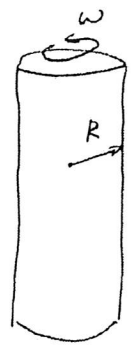
Problem 1



a) $\vec{j} = (0, 0, \sigma\omega R)\,\delta(r-R)$

We look for $\vec{A}$ in the form

$$\vec{A} = (0, 0, A_\varphi)$$

$$(\nabla^2 \vec{A})_\varphi = -\mu_0 \vec{j} = -\mu_0 \sigma\omega R\,\delta(r-R)\,\hat{\varphi}$$

$$(\nabla^2 \vec{A})_\varphi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial A_\varphi}{\partial r}\right) - \frac{A_\varphi}{r^2}$$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dA_\varphi}{dr}\right) - \frac{A_\varphi}{r^2} = -\mu_0 \sigma\omega R\,\delta(r-R)$$

General solution of the homogeneous equation

$$A_\varphi = ar + \frac{b}{r}$$

Inside $b=0$, outside $a=0$

$$A_\varphi(r) = \begin{cases} ar, & r<R \\ \frac{b}{r}, & r>R \end{cases}$$

Boundary conditions: $aR = \frac{b}{R}$

$$A_\varphi'(R+0) - A_\varphi'(R-0) = -\mu_0 \sigma\omega R$$

$$-\frac{b}{R^2} - a = -\mu_0 \sigma\omega R$$

$$a = \frac{\mu_0 \sigma\omega R}{2} \qquad b = \frac{\mu_0 \sigma\omega R^2}{2}$$

Field outside:

$$B_z = \frac{1}{r}\frac{\partial}{\partial r}(rA_\varphi) = 0$$

Field inside:

$$B_z = 2a = \mu_0 \sigma\omega R$$

From the boundary condition at $r=R$:

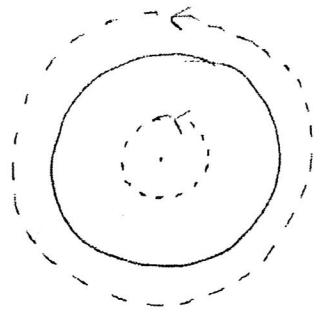
$$A_z'(R+0) - A_z'(R-0) = -\mu_0 B v$$

we find $a = -\mu_0 B v$

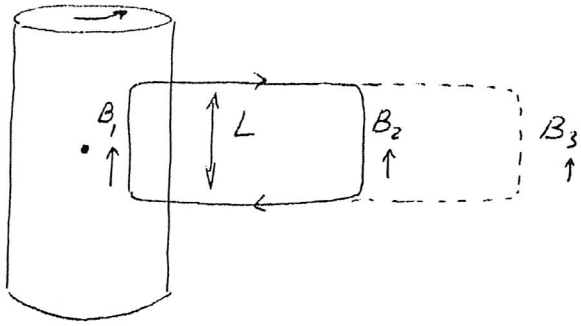
Since $A_z = A_z(r)$ only $B_\varphi \neq 0$:

$$B_\varphi = (\nabla \times \vec{A})_\varphi = -\frac{\partial A_z}{\partial r} = \frac{a}{r} = \frac{\mu_0 B v}{r}$$

Two contours C_1 and C_2 help explain the result from the Ampere's law



b) Consider the two contours



$$\oint \vec{B}\cdot d\vec{l} = \mu_0 j L$$

$$\Downarrow$$

$$B_1 L - B_2 L = \beta \omega R L \mu_0$$

and

$$B_1 L - B_3 L = \beta \omega R \mu_0$$

Subtracting one from the other we get $B_2 = B_3$.

Since a finite-radius cylinder cannot create constant at $r = \infty$ magnetic field $B_2 = B_3 = 0$

and $B_1 = \mu_0 \beta \omega R$.

From the fact that the contour can pass inside the cylinder at any distance to the axis the field B_1 must be constant.

c) $\vec{A} = (A_z, 0, 0)$ now as $j_z = \beta v \delta(r-R)$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dA_z}{dr}\right) = -\mu_0 \beta v \delta(r-R) \qquad A_z = A_z(r)$$

General solution is now

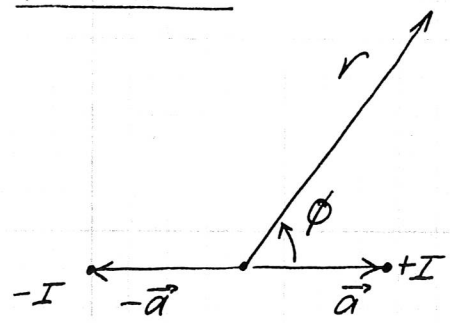
$$A_z(r) = a \ln r + b$$

b ← arbitrary constant which does not affect the field $\vec{B} = \nabla \times \vec{A}$ and can thus be set $= 0$

Inside $a = 0$

$$A_z(r) = \begin{cases} a \ln r, & r > R \\ 0, & r < R \end{cases}$$

Problem 2



Potential of a single wire $(+I)$

$$\vec{A} = \hat{z}\,\frac{\mu_0 I}{2\pi}\ln|\vec{r}-\vec{a}|$$

For both wires

$$\vec{A} = \hat{z}\,\frac{\mu_0 I}{2\pi}\ln\frac{|\vec{r}-\vec{a}|}{|\vec{r}+\vec{a}|}$$

for $r \gg a$, $|\vec{r}\pm\vec{a}| \approx r \pm \frac{\vec{a}\cdot\vec{r}}{r}$

and
$$\vec{A} = \hat{z}\,\frac{\mu_0 I}{2\pi}\ln\frac{1-\frac{\vec{a}\cdot\vec{r}}{r^2}}{1+\frac{\vec{a}\cdot\vec{r}}{r^2}} \approx \hat{z}\,\frac{\mu_0 I}{2\pi}\ln\left(1-\frac{2\vec{a}\cdot\vec{r}}{r^2}\right)$$

$$\approx \hat{z}\,\frac{\mu_0 I}{\pi}\,\frac{\vec{a}\cdot\vec{r}}{r^2} \qquad (1)$$

Magnetic field

$$\vec{B} = \nabla\times\vec{A} = -\frac{\mu_0 I}{2\pi}\,\hat{z}\times\nabla\ln\frac{|\vec{r}-\vec{a}|}{|\vec{r}+\vec{a}|}$$

Using that $\nabla\ln|\vec{r}-\vec{a}| = \frac{\nabla|\vec{r}-\vec{a}|}{|\vec{r}-\vec{a}|} = \frac{\vec{r}-\vec{a}}{(\vec{r}-\vec{a})^2}$

we get

$$\vec{B} = \frac{\mu_0 I}{2\pi}\left[\frac{(\vec{r}-\vec{a})\times\hat{z}}{(\vec{r}-\vec{a})^2} - \frac{(\vec{r}+\vec{a})\times\hat{z}}{(\vec{r}+\vec{a})^2}\right]$$

at large $r \gg a$

we use $\frac{1}{(\vec{r}\mp\vec{a})^2} \approx \frac{1}{r^2 \mp 2\vec{a}\cdot\vec{r}} \approx \frac{1}{r^2}\left(1 \pm \frac{2\vec{a}\cdot\vec{r}}{r^2}\right)$

to obtain

$$\vec{B} = \frac{\mu_0 I}{2\pi r^2}\left[(\vec{r}-\vec{a})\times\hat{z}\left(1+\frac{2\vec{a}\cdot\vec{r}}{r^2}\right)-(\vec{r}+\vec{a})\times\hat{z}\left(1-\frac{2\vec{a}\cdot\vec{r}}{r^2}\right)\right]$$

$$\approx \frac{\mu_0 I}{\pi r^2}\left(2\vec{r}\times\hat{z}\,\frac{\vec{a}\cdot\vec{r}}{r^2} - \vec{a}\times\hat{z}\right)$$

or $$\vec{B} = \frac{\mu_0 I}{\pi r^2}\left(2\,\vec{n}\times\hat{z}\,(\vec{a}\cdot\vec{n}) - \vec{a}\times\hat{z}\right)$$

where $\vec{n} = \vec{r}/r$.

The same result follows from taking a curl of the limiting expression (1).